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7.5 Coordinate Proofs for Quadrilaterals

On the coordinate plane, three algebraic tools let you prove exactly what kind of quadrilateral you have — no protractor needed. **Slope** ($m = \frac{y_2 - y_1}{x_2 - x_1}$) tests parallel sides (equal slopes) and perpendicular sides (slopes multiply to -1). **Distance** ($d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$) tests congruent sides or diagonals. **Midpoint** tests whether diagonals bisect each other. Chain these together to climb the hierarchy: both pairs of opposite sides parallel ⇒ parallelogram; add perpendicular adjacent sides (or congruent diagonals) ⇒ rectangle; add congruent sides (or perpendicular diagonals) ⇒ rhombus; both ⇒ square. Only one pair of parallel sides ⇒ trapezoid. Two distinct pairs of consecutive congruent sides (with perpendicular diagonals but no parallel sides) ⇒ kite. This worksheet uses full coordinates and asks you to justify the most specific classification possible.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Slopes $\frac{3}{4}$ and $\frac{3}{4}$. Parallel?

2. Slopes 4 and $-\frac{1}{4}$. Perpendicular?

3. Diagonals share midpoint $(2, 3)$. The figure must be a...

4. $A(0, 0)$, $B(6, 0)$. Find AB.

5. A parallelogram has four congruent sides. Name it.

6. A parallelogram has a right angle. Name it.

7. A quadrilateral has exactly one pair of parallel sides. Name it.

8. A rhombus has congruent diagonals. Name it.

9. A vertical side has slope...

10. $C(2, 3)$, $D(2, 8)$. Find CD.

11. Slopes $\frac{5}{2}$ and $\frac{2}{5}$. Perpendicular?

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12. In a rectangle, the diagonals are always…

13. To prove a quadrilateral is a parallelogram using only its diagonals, you show that they…

14. Vertices $A(0,0)$, $B(4,2)$, $C(6,6)$, $D(2,4)$. Classify $ABCD$ as specifically as possible and give the side length.

15. Vertices $W(0,4)$, $X(3,0)$, $Y(0,-5)$, $Z(-3,0)$. Show $WXYZ$ is a kite and give both distinct side lengths.

✓ Answer Key

1. Yes
2. Yes
3. Parallelogram
4. 6
5. Rhombus
6. Rectangle
7. Trapezoid
8. Square
9. Undefined
10. 5
11. No (product $\neq -1$)
12. Congruent
13. Bisect each other (share the same midpoint)
14. Rhombus (not a square); side $= 2\sqrt{5}$
15. Kite; $WX = ZW = 5, XY = YZ = \sqrt{34}$